

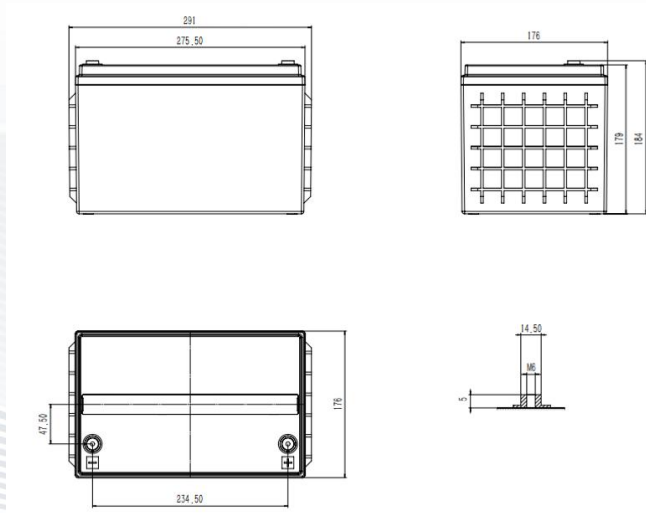
8XNFG90

Nickel-Zinc Battery



Dimension

(Unit: mm)



Application

- ☒ Data center
- ☒ Railway transportation
- ☒ Bank and financial market

Feature

- **High reliability:** Battery does not open circuit during discharge.
- **Ultimate safety:** Pass extreme tests (axe chopping, drilling, heavy crushing, fire exposure) with no explosion or fire risks. Safety performance far exceeds traditional batteries.
- **High discharge rate:** Supports discharge in the range of 45A-900A (0.5C-10C), balancing capacity and power requirements, flexibly adapting to different application scenarios.
- **Wide temperature adaptability:** Efficient and stable operation under -20° C to 55° C, reducing the environment requirement.
- **Eco-friendly:** >90% recyclability, made from non-polluting materials, promoting clean energy use and contributing to carbon footprint reduction.
- **High voltage platform:** Nominal voltage of 1.65V per cell, significantly higher than the 1.2V of Ni-MH and Ni-Cd batteries, offering greater output efficiency advantages.

Parameter

Parameter	8XNFG90
Rated voltage	13.2V (1.65V*8 cells)
Nominal capacity	90Ah (90A discharge to 10.4V at 25°C)
Nominal energy	1.188kWh
Weight	18.3 ± 0.3kg
Gravimetric energy density	79.2Wh/kg
Volumetric energy density	142.6Wh/L
Internal resistance	≤3mΩ
Open circuit voltage	≥14.40V (After 1 hour standard discharge)
Equalization charge voltage	15.20V, 0.3C (25°C)
Charging current	27A~90A
Standard discharge	90A
Operating temperature	Discharge: -20°C ~ 55°C Charge: 0°C ~ 40°C
Storage temperature	-20°C~+55°C
Self-discharge performance	1 month: 80% SOC, capacity retention ≥90%
	3 months: 80% SOC, capacity retention ≥80%
	6 months: 80% SOC, capacity retention ≥70%
Terminal type	M6 copper core
Case and cover material	V0 flame-retardant ABS engineering plastic
Cycle life (25°C)	500 cycles (1C 100%DOD) 800 cycles (1C 50%DOD)
Certification	
Case flame rating	UL 94 V-0
EU2023-1542	Passed
RoHS	Passed
UL9540A	In progress

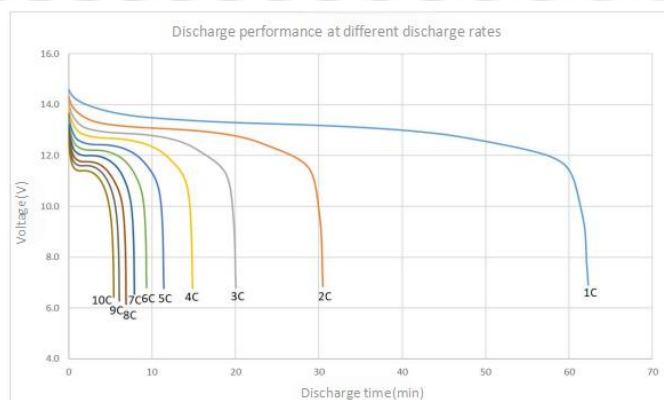
► Constant current discharge table (A/cell) (25°C)

Cut-off voltage	5m	10m	15m	20m	30m	45m	1h	2h	3h	4h	5h	6h	8h	10h
1.45V	621.8	419.7	306.3	234.8	161.6	111.2	85.3	43.8	29.5	22.2	17.8	14.8	11.1	8.9
1.40V	694.4	437.4	314.8	243.0	166.0	113.4	86.6	44.0	29.6	22.3	17.9	14.9	11.2	9.0
1.35V	743.6	458.0	322.9	247.0	168.0	114.2	86.8	44.1	29.7	22.4	18.0	15.0	11.2	9.0
1.30V	784.7	475.6	330.2	249.4	169.1	114.6	87.0	44.1	29.7	22.4	18.0	15.1	11.3	9.0
1.25V	823.0	489.8	334.5	251.3	170.2	115.3	87.4	44.2	29.8	22.5	18.1	15.1	11.4	9.1
1.20V	870.9	497.6	341.7	253.6	171.9	116.6	88.5	44.4	30.0	22.6	18.2	15.2	11.4	9.2

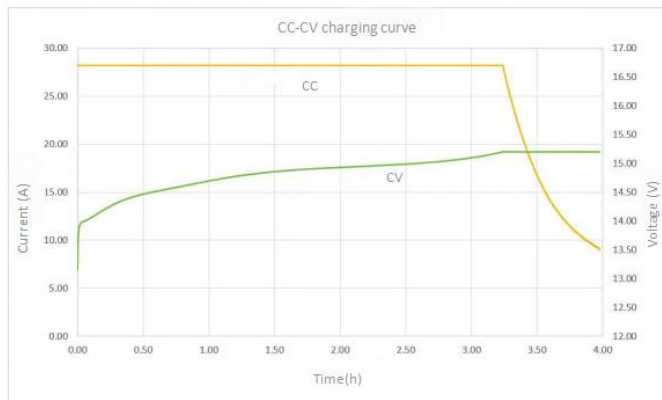
► Constant power discharge table (W/cell) (25°C)

Cut-off voltage	5m	10m	15m	20m	30m	45m	1h	2h	3h	4h	5h	6h	8h	10h
1.45V	815.1	614.5	484.8	385.4	261.8	179.9	140.9	72.8	48.7	36.7	29.5	24.8	18.7	15.1
1.40V	907.4	641.2	493.4	392.6	272.9	185.2	141.4	73.5	49.3	37.0	29.6	24.9	18.9	15.2
1.35V	973.8	663.7	502.0	397.2	274.3	186.7	141.6	73.9	49.4	37.2	29.9	25.1	19.0	15.3
1.30V	1007.0	683.5	509.9	398.1	274.9	188.7	141.6	74.2	49.6	37.4	30.0	25.2	19.0	15.3
1.25V	1053.8	695.4	512.7	399.4	276.1	189.1	141.8	74.4	49.7	37.5	30.1	25.2	19.1	15.3
1.20V	1096.2	703.6	513.9	400.6	278.1	190.2	142.7	74.6	49.8	37.5	30.1	25.3	19.1	15.3

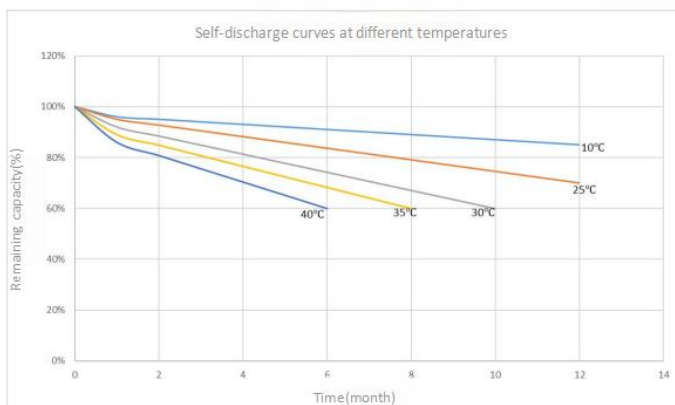
Discharge performance at different discharge rates



CC-CV charge curve



Self-discharge curves at different temperatures



Discharge capacity at different temperatures

